

Color LightField: Estimation Of View-point Dependant Color Dispersion In Waveguide Display

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Figure 1: Left: CGI of a virtual Utah Teapot; Middle: A see-through photo of a Utah Teapot that captured from a randomly picked perspective from the light field; Right: A light field acquisition rig utilising a 3D printer as a 2D motorized stage.

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1 INTRODUCTION

Diffraction waveguides have become the mainstream optical combiners for commercial augmented reality (AR) near-eye displays (NED) such as Microsoft HoloLens 1 & 2, Magic Leap One and Snap Spectacles due to their compactness and high transmission rate. The waveguide display projects the image onto the user's eyes by coupling the collimated projector light into the waveguide substrate to propagate forward via total internal reflection (TIR). When the light reaches the exit pupil expander (EPE), it is decoupled through a diffractive optical element (DOE), allowing the light to exit the system to form an image to the user. However, the diffractive grating on the optical element introduces undesired "rainbow effect" into the user's view which affects the viewing experience negatively. The "rainbow effect" may be caused by a number of factors, such as light sources in the real environment and the field sequential color method. Our research method in this paper only focuses on the factors within the system.

The diffractive grating is sensitive and selective to a very narrow spectrum of light. Therefore in order to display full color, 3 DOEs are stacked sequentially, reflecting red, green and blue color. The stacked DOEs in this method each slightly diffract the light coming from the others which cause a heavy color cross-talk in the image and thus resulting in the visible "rainbow effect".

The diffraction efficiency in waveguide display can be simulated in software such as ZEMAX but the color dispersion may still persist in mass production processes due to small flaws in the yields that are difficult to quantify and detect. Moreover, the color dispersion is viewpoint dependent which means the dispersion pattern that the user sees varies depending on the position of the user's eye relative to the EPE within the eyebox. When wearing a AR NED, the position of the eyes vary person to person because of difference in interpupillary distance, nose height etc. Therefore, a calibration method that can quantify and detect the color at the final output of the waveguide at any given viewpoint is required for accurate improvement of color dispersion. Srikanth et al. [Sridharan et al. 2013] proposed a color correction method based on color blending but it is not dynamic to address the viewpoint-dependent color dispersion problem. Itoh et al. proposed the 4D to 4D mapping light field based calibration method to model and correct spatial distortions in AR NEDs [Itoh and Klinker 2015], and in [Itoh et al. 2016], a PSF model is proposed to compensate blurriness but not for the chromatic aspect.

In this paper, we propose a 2-step method to estimate the color dispersion pattern by capturing and generating a light field image within the eyebox. And then we determine the perspective view of the user by matching the user eye position in the eye box by using eye tracking. We believe our method can be applied to all types of diffractive waveguide based display such as surface relief

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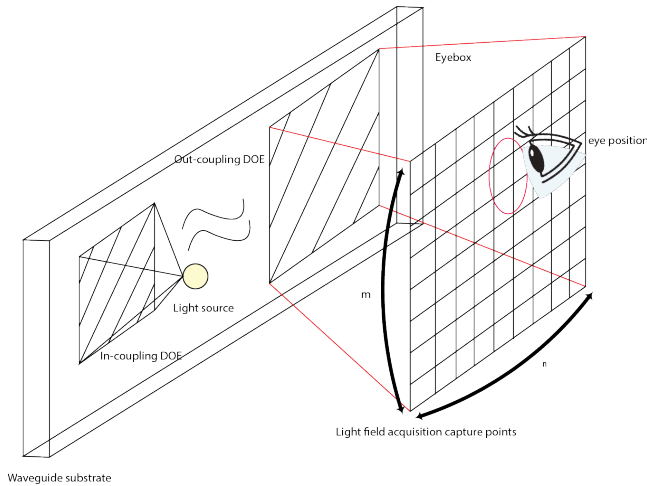


Figure 2

grating(SRG) and holographic volume grating(HVG). Our method can also be further improved to include external stray light in the future by capturing the light field of the display output in a synced light stage.

2 METHOD OVERVIEW

There are 2 steps in our proposed method: acquiring a light field image and creating an observer from the eye position's perspective, in order to estimate what the user actually sees. Fig 2 shows the basic concept of our method. We capture the light field images in a time sequential manner as the color dispersion does not change over time but rather spatially. Here we describe the setup that we used to capture the images for the light field computation. As we are capturing the light field linearly, a 2D rail or stage that can move the camera along the horizontal and vertical axes in a fixed interval manner is required. For this we program a 3D printer such that the printing head moves systematically in the XZ-axis within the eyepiece of the AR NED. Then, we 3D printed a mount for the camera which in our case is a smartphone case to fixate the camera to the printing head. A neutral density filter is used to allow an optimal entrance pupil of at least 3 mm in the camera. We then placed the AR NED in front of the camera at eye relief distance as shown in Fig 1(right). With this setup, we capture 50x40 images with 1 mm intervals in both X and Z axes under consistent camera settings across the capture points. Then the images are processed through the light field computation pipeline as described in [Unger et al. 2003].

Next, we acquire the position of the eye by the method proposed by Idris et al.[Idris et al. 2018] by measuring the pupil or the iris center. Though in recent AR NEDs with eye tracker feature such as HoloLens 2, the eye tracking data that approximate the eye position can be obtained by running the calibration application. After the eye position is known, we can then create a virtual observer that output a specific light field perspective that corresponds to what the user sees.

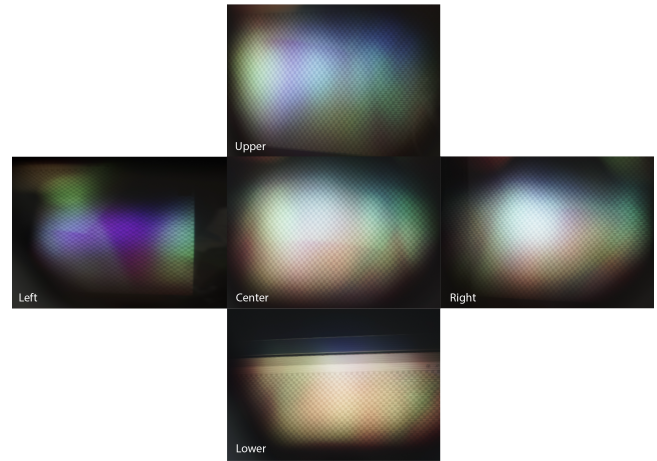


Figure 3: Estimated color dispersion pattern at randomly picked perspective shown at upper; right; center; left; lower side of the eyepiece.

3 PRELIMINARY RESULT

In our experimental setup, we used Microsoft HoloLens 2 as our target device, which, in our opinion, has a more evident color dispersion problem compared to others. We compute the images captured within the eyepiece of AR NED into a light field image. In Fig 3, we show the perspective views when the eye is at the upper, right, center, and lower side of the eyepiece. We can observe that the color dispersion varies in chromatic colors as well as the dispersion pattern spatially.

4 FUTURE WORKS

In this paper, we describe the method on how to estimate the color dispersion pattern that the user sees by using light field and eye tracking technique. We hope that this work can be a foundation for upcoming color correction and color dispersion reduction methods in the next generation waveguide based AR NEDs.

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